Wenxi Wang

Assistant Professor
Department of Computer Science
The University of Virginia

wenxiw@virginia.edu
wenxiwang.github.io

Research Interests

My research focuses on advancing Neuro-Symbolic AI to improve the scalability and efficiency of automated reasoning systems, while enabling AI models to develop reasoning and verification capabilities. Beyond foundational research, I design specialized Neuro-Symbolic methods to enhance software reliability, including modern AI systems, and explore verifiable code generation to make software development more trustworthy and error-resistant.

Employment

2024–Present **Assistant Professor**
The University of Virginia (UVA), Department of Computer Science

Education

2024 **Doctor of Philosophy**, *The University of Texas at Austin (UT Austin)*
Research Areas: Software Engineering, Formal Methods, Machine Learning
Advisor: Sarfraz Khurshid

2017 **Master of Philosophy**, *The University of Melbourne (UoM)*
Research Areas: Automated Logical Reasoning
Thesis: A Bit-Vector Solver Based on Word-Level Propagation
Advisor: Peter J. Stuckey and Harald Sondergaard

2014 **Bachelor of Engineering**, *Dalian University of Technology (DUT)*
Major: Computer Science and Technology
Advisor: Yanming Shen

Publications

[1] Lize Shao*[1], Michael Cardei*, Zichen Xie, Ferdinando Fioretto, **Wenxi Wang**. *"Constrained Code Generation with Discrete Diffusion"*, In *The Fortieth Annual Conference on Neural Information Processing Systems (Neurips 2026, Main Track)*. **[PDF]**

[2] Zichen Xie, Mrigank Pawagi, Yuxin Liu, Aaditi Rai, Lize Shao, John Berberian Jr., Sicong Che, **Wenxi Wang**. *"VeriContest: A Competitive-Programming Benchmark for Verifiable Code Generation"*, In *The Fortieth Annual Conference on Neural Information Processing Systems (Neurips 2026, Evaluation and Dataset Track)*. **[PDF]**

[3] Zichen Xie, **Wenxi Wang**. *"Can LLMs Reason Like Automated Theorem Provers for Rust Verification? VCoT-Bench: Evaluating via Verification Chain of Thought"* In *The forty-third International Conference on Machine Learning (ICML 2026)*. **[PDF]**

[1]* denotes that these authors contribute equally to the paper.

[4] Arjun Tandon, Mehmet Fırat Dündar, Milkiyas Gebremichael Gebru, Darko Marinov, Yiling Lou, **Wenxi Wang**. *"Re-evaluating Detection of Equivalent Mutants Using LLMs: We Should Properly Measure How Far We Are"* In *The 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. 🎗 **ACM SIGSOFT Distinguished Paper Award**. **[PDF]**

[5] Chaitanya Shahane, Derek Hansen, **Wenxi Wang**, Pengyu Nie *"Differential Inline Testing: Framework, Test Generation, and Application"* In *The 36th IEEE International Conference on Collaborative Advances in Software and Computing (CASCON 2026)* (to appear).

[6] Mrigank Pawagi, Lize Shao, Hyeonmin Lee, Yixin Sun, **Wenxi Wang**. *"RFCScope: Detecting Logical Ambiguities in Internet Protocol Specifications"* In *The 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025)*. **[PDF]**

[7] **Wenxi Wang**, Yang Hu, Mohit Tiwari, Sarfraz Khurshid, Kenneth L. McMillan, Risto Miikkulainen. *"NeuroBack: Improving CDCL SAT Solving using Graph Neural Networks."* In *The 12th International Conference on Learning Representations (ICLR 2024)*. **[PDF]**

[8] Yang Hu*, **Wenxi Wang***, Sarfraz Khurshid, Kenneth L. McMillan, Mohit Tiwari. *"Fixing Privilege Escalations in Cloud Access Control with MaxSAT and Graph Neural Networks."* In *The 38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*. **[PDF]**

[9] Armin Biere, Nils Froleyks, **Wenxi Wang**. *"CadiBack: Extracting Backbones with CaDiCal."* In *The 26th International Conference on Theory and Applications of Satisfiability Testing (SAT 2023)*. Tool Paper. **[PDF]**

[10] **Wenxi Wang**, Yang Hu, Kenneth L. McMillan, Sarfraz Khurshid. *"SymMC: Approximate Model Enumeration and Counting Using Symmetry Information for Alloy Specifications."* In *The 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2022)*. **[PDF]**

[11] Chengpeng Li, Chenguang Zhu, **Wenxi Wang**, August Shi. *"Repairing Order-Dependent Flaky Tests via Test Generation."* In *The 44th International Conference on Software Engineering (ICSE 2022)*. **[PDF]**

[12] **Wenxi Wang**, Pu Yi, Sarfraz Khurshid, Darko Marinov. *"Initial Results on Counting Test Orders for Order-Dependent Flaky Tests using Alloy."* In *The 33rd IFIP International Conference on Testing Software and Systems (ICTSS 2021)*. Note: Short Paper. **[PDF]**

[13] Yang Hu, **Wenxi Wang**, Casen Hunger, Riley Wood, Sarfraz Khurshid, Mohit Tiwari. *"ACHyb: A Hybrid Analysis Approach to Detect Kernel Access Control Vulnerabilities."* In *The 29th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2021)*. **[PDF]**

[14] Jiayi Yang, **Wenxi Wang**, Darko Marinov, Sarfraz Khurshid. *"AlloyMC: Alloy Meets Model Counting."* In *The 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2020)*. Tool Demo. **[PDF]**

[15] Muhammad Usman, **Wenxi Wang**, Sarfraz Khurshid. *"TestMC: Testing Model Counters using Differential and Metamorphic Testing."* In *The 35th IEEE/ACM International Conference on Automated Software Engineering (ASE 2020)*. **[PDF]**

[16] **Wenxi Wang**, Muhammad Usman, Alyas Almaawi, Kaiyuan Wang, Kuldeep S. Meel, Sarfraz Khurshid. *"A Study of Symmetry Breaking Predicates and Model Counting."* In *International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2020)*. **[PDF]**

[17] Muhammad Usman, **Wenxi Wang**, Kaiyuan Wang, Cagdas Yelen, Nima Dini, Sarfraz Khurshid. *"A Study of Learning Likely Data Structure Properties using Machine Learning Models."* In *International Journal on Software Tools for Technology Transfer (STTT 2020)*. **[PDF]**

[18] Muhammad Usman, **Wenxi Wang**, Kaiyuan Wang, Marko Vasic, Haris Vikalo, Sarfraz Khurshid. *"A Study of the Learnability of Relational Properties (Model Counting Meets Machine Learning)."* In *The 41st ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI 2020)*. **[PDF]**

[19] Muhammad Usman, **Wenxi Wang**, Kaiyuan Wang, Cagdas Yelen, Nima Dini, Sarfraz Khurshid. *"A Study of Learning Data Structure Invariants Using Off-the-shelf Tools."* In *The 26th International SPIN Symposium on Model Checking of Software (SPIN 2019)*. **[PDF]**

[20] **Wenxi Wang**, Kaiyuan Wang, Milos Gligoric, Sarfraz Khurshid. *"Incremental Analysis of Evolving Alloy Models."* In *International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2019)*. **[PDF]**

[21] **Wenxi Wang**, Kaiyuan Wang, Mengshi Zhang, Sarfraz Khurshid. *"Learning to Optimize the Alloy Analyzer."* In *The 12th IEEE International Conference on Software Testing, Verification and Validation (ICST 2019)*. **[PDF]**

[22] **Wenxi Wang**, Harald Sondergaard, Peter J. Stuckey. *"Wombit: A Portfolio Bit-Vector Solver using Word-Level Propagation."* In *Journal of Automated Reasoning (JAR 2018)*. **[PDF]**

[23] **Wenxi Wang**, Harald Sondergaard, Peter J. Stuckey. *"A Bit-Vector Solver with Word-Level Propagation."* In *Integration of AI and OR Techniques in Constraint Programming (CPAIOR 2016)*. **[PDF]**

Paper Submissions & Preprints

[24] Zichen Xie*, Mrigank Pawagi*, Lize Shao*, Yang Hu, **Wenxi Wang**. *"Detecting Inconsistencies in Model Specifications with LLM-as-Verifier Reasoning" arXiv preprint arXiv:2610.01847*, 2023. **[PDF]**

Patents

[25] Amit Goel, Dejan Jovanovic, Neha Rungta, **Wenxi Wang**. *"Optimizing SMT problem encoding for application-specific workloads with machine learning." U.S. Patent Application, Pending*, 2023.

Internships

5/2022–8/2022 **Applied Scientist Intern**, *Automated Reasoning Group*, Amazon Web Services
Host: Dejan Jovanovic
Project: Optimizing SMT problem encoding for application-specific workloads with Graph Neural Networks

5/2019–8/2019 **Research Intern**, *Software Quality & Security Lab*, Fujitsu Research of America
Host: Hiroaki Yoshida
Project: Automated program repairs for static analysis violations

9/2017–8/2018 **Research Intern**, *Department of Computing*, Hong Kong Polytechnic University
Host: Max Yu Pei
Project: Mutation-based fault localization with minimal unsatisfiable core analysis

Awards and Scholarships

2026 ACM SIGSOFT Distinguished Paper Award at ISSTA'26 [4]
2026 Silver Reviewer Award at ICML'26
2023–2024 George J. Heuer, Jr. Ph.D. Endowed Graduate Fellowship, UT Austin
2022 MIT EECS Rising Stars
2014–2016 Melbourne International Research Scholarship, UoM
2014–2016 Melbourne International Fee Remission Scholarship, UoM
2014 Province Excellent Graduates Award, Liaoning Province, China
2012–2013 China National Scholarship, Ministry of Education of China

Teaching

Fall 2026 Machine Learning for Software Reliability (CS6501), graduate Level, UVA
Fall 2025 Machine Learning for Software Reliability (CS6501), graduate Level, UVA
Spring 2025 Software Testing (CS3250), undergraduate Level, UVA
Fall 2024 Machine Learning for Software Reliability (CS6501), graduate Level, UVA

Mentoring

PhD Zichen Xie (Fall 2025–Present, UVA, co-authored papers [3], [2])
PhD Lize Shao (Fall 2025–Present, UVA, co-authored papers [6], [2], [1])
PhD Xiaoyan Guo (Fall 2026–Present, UVA)
Rotation Rishov Paul (2024 Fall Semester, UVA)
Master Ao Guo (Summer 2026–Present, UVA)
Master Derek Joseph Hansen (Fall 2025, UVA, co-authored paper [5])
Master John Edwin Berberian (Fall 2025, UVA, co-authored paper [2])
Master Chaitanya Rajendra Shahane (Fall 2024–Fall 2025, UVA, co-authored paper [5])
Master Sicong Che (Fall 2023–Fall 2025, UT Austin, co-authored paper [2])
Undergraduate Aaditi Rai (Fall 2025, UVA, co-authored paper [2])
Undergraduate Carter Opperman (Fall 2024–Present, UVA)
Internship Arjun Tandon (Fall 2025–present, Indraprastha Institute of Information Technology Delhi, co-authored paper [4])

Internship	Mrigank Pawagi (Fall 2024–Summer 2025, Indian Institute of Science, co-authored paper [6])
Internship	Tianyi Huang (Fall 2024–Spring 2025, University of Illinois Urbana-Champaign)
Internship	Yuxin Liu (Fall 2025, Rice University, co-authored paper [2])
Internship	Zhonghan Wang (2024–2025, Chinese Academy of Sciences)
Master	Sicong Che (2022–2025, UT Austin, co-authored paper [2])
Master	Jiayi Yang (2019–2024, UT Austin, co-authored papers [14])
Ph.D. Student	Muhammad Usman (2019–2021, UT Austin, co-authored papers [15, 16, 17, 18, 19])

Committee Services

Program Committee (PC):

OOPSLA 2027	PC member in ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)
FSE 2027	PC member in International Conference on the Foundations of Software Engineering (FSE)
ICSE 2027	PC member in International Conference on Software Engineering
ICSE 2026	PC member in International Conference on Software Engineering
KDD 2026	Reviewer in ACM SIGKDD Conference on Knowledge Discovery and Data Mining
CAV 2025	PC member in International Conference on Computer Aided Verification
OOPSLA 2025	PC member in Conference on Object-Oriented Programming Systems, Languages, and Applications
LLM4Code 2025 Workshop	PC member in International Workshop on Large Language Models for Code
ASE 2024	Session chair of "LLM for SE 2" and "SE for AI 2" sessions, and PC member in IEEE/ACM International Conference on Automated Software Engineering
ICML 2024	PC member in International Conference on Machine Learning
ICLR 2024	PC member in International Conference on Learning Representations
TOSEM 2023	Reviewer in ACM Transactions on Software Engineering and Methodology
ECOOP 2023	Session chair of the "Verification and Testing" session and extended PC member in European Conference on Object-Oriented Programming
NeurIPS 2023	PC member in Conference on Neural Information Processing Systems
ICML 2023	PC member in International Conference on Machine Learning
NeurIPS 2022	PC member in Conference on Neural Information Processing Systems

NSF Proposal Review Panel:

NSF CCF 2025	Proposal Review Panelist, Software and Hardware Foundations (SHF) Program

Artifact Evaluation Committee (AEC):

PLDI 2023	AEC member in Conference on Programming Language Design and Implementation
ISSTA 2023	AEC member in International Symposium on Software Testing and Analysis
ECOOP 2023	AEC member in European Conference on Object-Oriented Programming
USENIX SEC 2023	AEC member in USENIX Security Symposium
ISSTA 2022	AEC member in International Symposium on Software Testing and Analysis

PLDI 2022	AEC member in Conference on Programming Language Design and Implementation
PLDI 2021	AEC member in Conference on Programming Language Design and Implementation

Professional Services

Web Committee Member	Department of Computer Science, UVA
PhD Dissertation Committee Member	for PhD student Mingtian Tan, Department of Computer Science, UVA